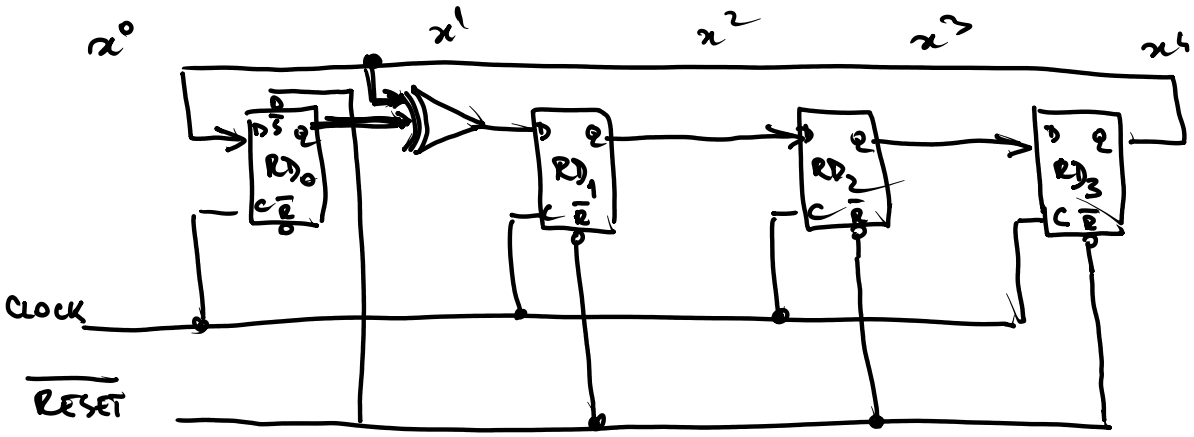
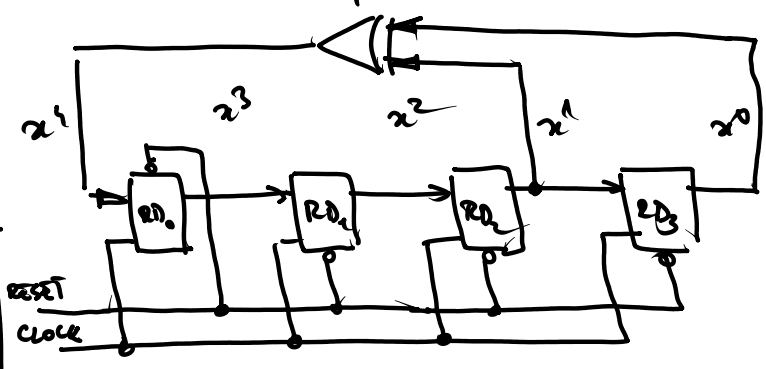
$G(x) = x^4 + x + 1$ Feedback register

$\gcd(G(x)) = 4 \Rightarrow$ sequence length $2^4 - 1$



LSB RD_0	RD_1	RD_2	RD_3 MSB	Dec
1	0	0	0	1
0	1	0	0	2
0	0	1	0	4
0	0	0	1	8
1	1	0	0	3
0	1	1	0	6
0	0	1	1	12
1	1	0	1	11
1	0	1	0	5
0	1	0	1	10
1	1	1	0	7
0	1	1	1	14
1	1	1	1	15
1	0	1	1	13
1	0	0	1	9
1	0	0	0	1

External XOR feedback

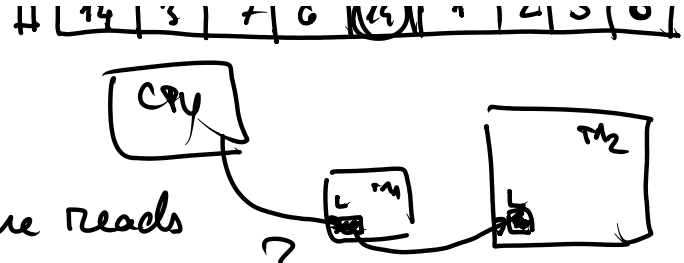


b) LRU - Age register

14, 7, 6, 12, 10, 6, 7, 14, 7, 3, 14

index N 4-way SA

	Code	0	1	2	3	0*	1*	2*	3*
		0	0	0	0	0	0	0	0
M	14	14	0	0	0	0	0	0	0
M	7	14	7	0	0	1	0	0	0
M	6	14	7	6	0	2	1	0	0
M	12	14	7	6	12	3	2	1	0
M	10	10	7	6	12	0	3	2	1
H	6	10	7	6	12	1	3	0	2
H	7	10	7	6	12	2	0	1	3
M	14	10	7	6	14	3	1	2	0
H	7	10	7	6	14	3	0	2	1
M	3	3	7	6	14	0	1	3	2
H	14	3	7	6	14	1	2	3	0



3.4 Write policies

Coarsely, ≈ 90% of all references are reads
only 10% writes

?

Cache coherency

For reads, in the case of misses → Allocate (locality principle)

For writes (hits)
- a) Write Through (write in both M1 and M2)
  - performance issues mitigated w/ a write buffer
- b) Write Back (write only in M1)
  - Coherence at replacement (write in M2 only at replacement) - block update
  - dirty bit

For write misses
- α) Write No Allocate
- β) Write Allocate

From a practical standpoint
- Write Through + Write No Allocate
- Write Back + Write Allocate

a + α - write in M2
b + β - write in M1 first